

Create and mount an Amazon EFS file system

using Amazon EC2 Launch Instance Wizard

Step 1: Create the security groups

For this step, you create two security groups - one that is attached to the Amazon EC2 instance and one that is attached to the Amazon EFS file system.

Open the [Amazon VPC dashboard](#) and In the left navigation pane, choose Security Groups. There will be a default security group listed. Create the following two additional security groups in the default VPC. For detailed steps, see [Creating a Security Group](#).

- EC2-sg: This security group will be attached to the EC2 instance and it allows only SSH connection inbound to the EC2 instance and any outbound connectivity.
- EFS-sg: This security group will be attached to the EFS file system and allows only TCP connection on port 2409 from the EC2 instance and any outbound connectivity.

Security Group1: -

The image shows two screenshots from the AWS Management Console. The top screenshot is the 'Security Groups (1)' page in the VPC dashboard. It shows a table with one entry: 'default' security group with ID 'sg-094372c1590200680' and VPC ID 'vpc-0e9096f0008e45be5'. An arrow points to the 'Create security group' button in the top right. The bottom screenshot is the 'Create security group' wizard. It has three sections: 'Basic details', 'VPC', and 'Tags'. In the 'Basic details' section, the 'Security group name' is 'EC2-sg' and the 'Description' is 'allows only SSH connection'. In the 'VPC' section, the 'VPC' is 'vpc-0e9096f0008e45be5' (default). An arrow points to the 'Create' button at the bottom right.

Security Groups (1)

Name	Security group ID	Security group name	VPC ID	Description
-	sg-094372c1590200680	default	vpc-0e9096f0008e45be5	default VPC security group

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name **EC2-sg**
Name cannot be edited after creation.

Description **allows only SSH connection**

VPC

vpc-0e9096f0008e45be5
vpc-0e9096f0008e45be5 (default) ✓

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional
SSH	TCP	22	Anywhere... 0.0.0.0/0	

[Add rule](#)

Outbound rules [Info](#)

Type	Protocol	Port range	Destination	Description - optional
All traffic	All	All	Custom 0.0.0.0/0	

[Add rule](#)

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags

[Cancel](#) [Create security group](#)

Security Group2: -

Security Groups (1) [Info](#)

[Find resources by attribute or tag](#)

Name	Security group ID	Security group name	VPC ID	Description
-	sg-094372c1590200680	default	vpc-0e9096f0008e45be5	default VPC security group

[Create security group](#)

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

EPS-sg

Description [Info](#)

allows only TCP connection on port 2409 from the EC2 instance

VPC [Info](#)

vpc-0e9096f0008e45be5

(default) ☒

Select EC2-sg security group

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
NFS	TCP	2049	Custom sg-0f0a1f42310af0e4c	
Add rule				

Outbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info
All traffic	All	All	Custom 0.0.0.0/0	
Add rule				

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)
You can add up to 50 more tags.

[Cancel](#) [Create security group](#)

VPC dashboard [x](#)

EC2 Global View [🔗](#)

Filter by VPC [▼](#)

▼ Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only Internet gateways

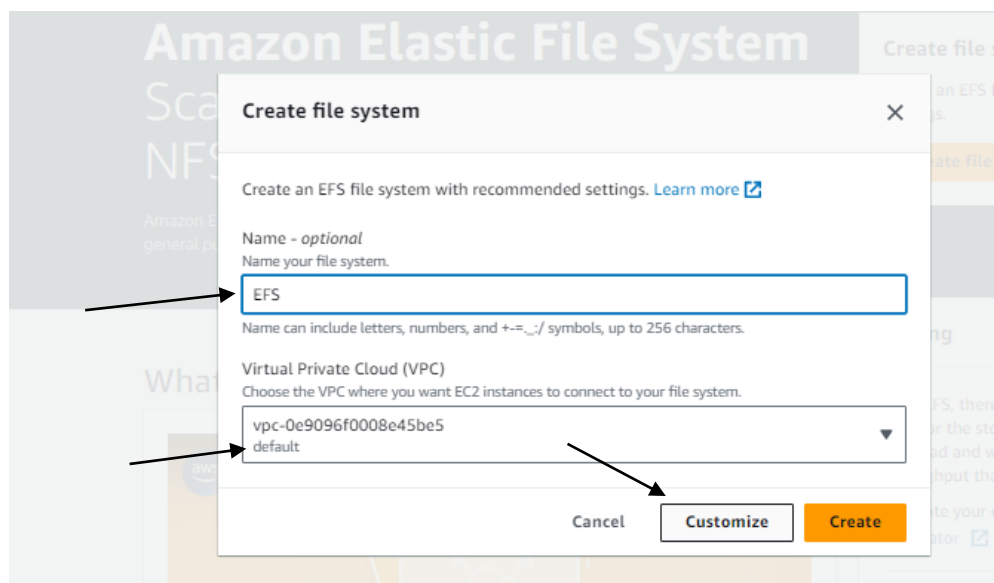
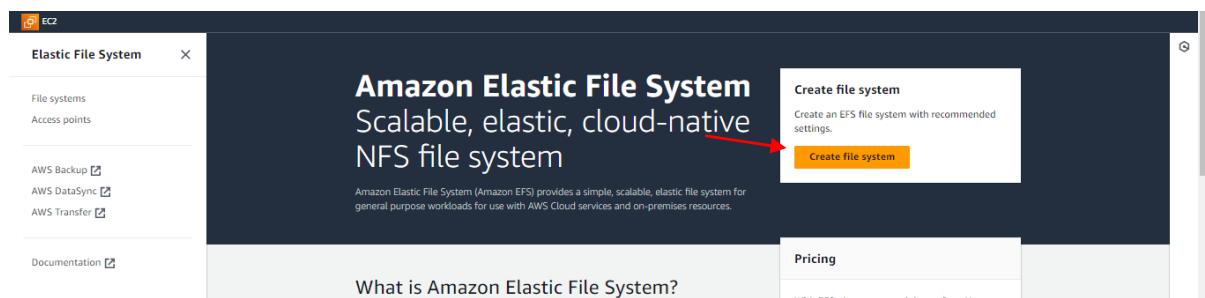
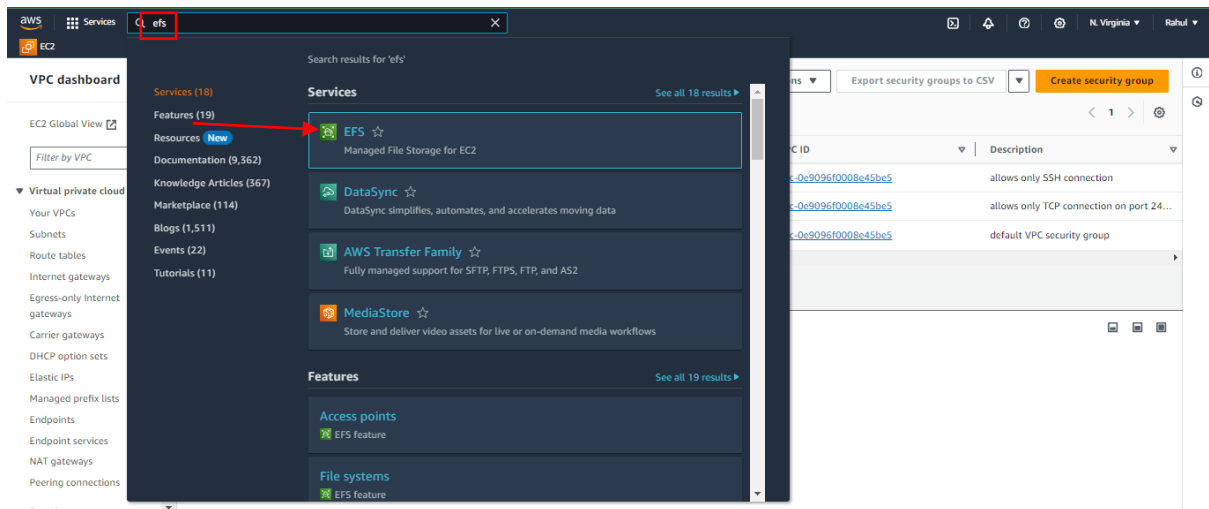
Security Groups (5) [Info](#)

[🔄](#) [Actions](#) [Export security groups to CSV](#) [Create security group](#)

☐	Name	Security group ID	Security group name	VPC ID	Description
☐	-	sg-0f0a1f42310af0e4c	EC2-sg	vpc-0e9096f0008e45be5	allows only SSH connection
☐	-	sg-0cb867ff79f4782e	EFS-sg	vpc-0e9096f0008e45be5	allows only TCP connection on port 24...
☐	-	sg-094372c1590200680	default	vpc-0e9096f0008e45be5	default VPC security group

Step 2: Create the EFS file system

Navigate to [Amazon EFS in the AWS Management Console](#) and choose Create file system.



Amazon EFS > File systems > Create

Step 1
File system settings

Step 2
Network access

Step 3 - optional
File system policy

Step 4
Review and create

File system settings

General

Name - optional
Name your file system.
EFS

File system type
Choose to either store data across multiple Availability Zones or within a single Availability Zone. [Learn more](#)

☒ **Regional**
Offers the highest levels of availability and durability by storing file system data across multiple Availability Zones within an AWS Region.

☐ **One Zone**
Provides continuous availability to data within a single Availability Zone within an AWS Region.

Automatic backups
Automatically backup your file system data with AWS Backup using recommended settings. Additional pricing applies. [Learn more](#)

☐ Enable automatic backups

We recommend that you create a backup policy for your file system

Step 1
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Review and create

Lifecycle management

Automatically save money as access patterns change by moving files into the Infrequent Access (IA) or Archive storage class. [Learn more](#)

Transition into Infrequent Access (IA)
Transition files to IA based on the time since they were last accessed in Standard storage.

None

Transition into Archive
Transition files to Archive based on the time since they were last accessed in Standard storage.

None

Transition into Standard
Transition files back to Standard storage based on when they are first accessed in IA or Archive storage.
Valid only when you have enabled lifecycle management to transition files into IA or Archive storage.

None

Encryption
Choose to enable encryption of your file system's data at rest. Uses the AWS KMS service key (aws/elasticfilesystem) by default. [Learn more](#)

☐ Enable encryption of data at rest

Performance settings

Throughput mode
Choose a method for your file system's throughput limits. [Learn more](#)

☐ **Enhanced**
Provides more flexibility and higher throughput levels for workloads with a range of performance requirements.

☒ **Bursting**
Provides throughput that scales with the amount of storage for workloads with basic performance requirements.

► Additional settings

Step 4
Review and create

default

Mount targets

A mount target provides an NFSv4 endpoint at which you can mount an Amazon EFS file system. We recommend creating one mount target per Availability Zone. [Learn more](#)

Availability zone	Subnet ID	IP address	Security groups
us-east-1a	subnet-04d8a871409cf6025	Automatic	Choose security groups sg-094372c1590200680 default
us-east-1b	subnet-072eadad91a5396d9	Automatic	Choose security groups sg-094372c1590200680 default
us-east-1c	subnet-0582542460d795f57	Automatic	Choose security groups sg-094372c1590200680 default
us-east-1d	subnet-087a8a59564a27f3a	Automatic	Choose security groups sg-094372c1590200680 default
us-east-1f	subnet-06f253eef3a13dc5e	Automatic	Choose security groups sg-094372c1590200680 default

In the Security groups column, delete the existing security groups and add the EFS-sg security group.

Step 2

Network access

Step 3 - optional

File system policy

Step 4

Review and create

Virtual Private Cloud (VPC)

Learn more

Choose the VPC where you want EC2 instances to connect to your file system.

vpc-0e9096f0008e45bc5

default

Mount targets

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Availability zone	Subnet ID	IP address	Security groups	
us-east-1a	subnet-04d8a871409cf6025	Automatic	Choose security groups	Remove
us-east-1b	subnet-072eadad91a5396d9	Automatic	Choose security groups	Remove
us-east-1c	subnet-0582542460d795f57	Automatic	Choose security groups	Remove
us-east-1d	subnet-087a8a59564a27f3a	Automatic	Choose security groups	Remove
us-east-1f	subnet-06f253eef3a13de5e	Automatic	Choose security groups	Remove

Add mount target

File system settings

Step 2

Network access

Step 3 - optional

File system policy

Step 4

Review and create

Virtual Private Cloud (VPC)

Learn more

Choose the VPC where you want EC2 instances to connect to your file system.

vpc-0e9096f0008e45bc5

default

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us-east-1a	subnet-04d8a871409cf6025	Automatic	Choose security groups	Remove
us-east-1b	subnet-072eadad91a5396d9	Automatic	Choose security groups	Remove
us-east-1c	subnet-0582542460d795f57	Automatic	Choose security groups	Remove
us-east-1d	subnet-087a8a59564a27f3a	Automatic	Choose security groups	Remove
us-east-1f	subnet-06f253eef3a13de5e	Automatic	Choose security groups	Remove

Add mount target

A mount target provides an NFSv4 endpoint at which you can mount an Amazon EFS file system. We recommend creating one mount target per Availability Zone. [Learn more](#)

Availability zone	Subnet ID	IP address	Security groups	
us-east-1a	subnet-04d8a871409cf6025	Automatic	Choose security groups	Remove
us-east-1b	subnet-072eadad91a5396d9	Automatic	Choose security groups	Remove
us-east-1c	subnet-0582542460d795f57	Automatic	Choose security groups	Remove
us-east-1d	subnet-087a8a59564a27f3a	Automatic	Choose security groups	Remove
us-east-1f	subnet-06f253eef3a13de5e	Automatic	Choose security groups	Remove

Add mount target

Amazon EFS > File systems > Create

Step 1

File system settings

Step 2

Network access

Step 3 - optional

File system policy

Step 4

Review and create

File system policy - optional

Policy options

Select one or more of these common policy options, or create a custom policy using the editor. [Learn more](#)

☐ Prevent root access by default*
 ☐ Enforce read-only access by default*
 ☐ Prevent anonymous access
 ☐ Enforce in-transit encryption for all clients

* Identity-based policies can override these default permissions.

Grant additional permissions

Policy editor (JSON)

Clear

1

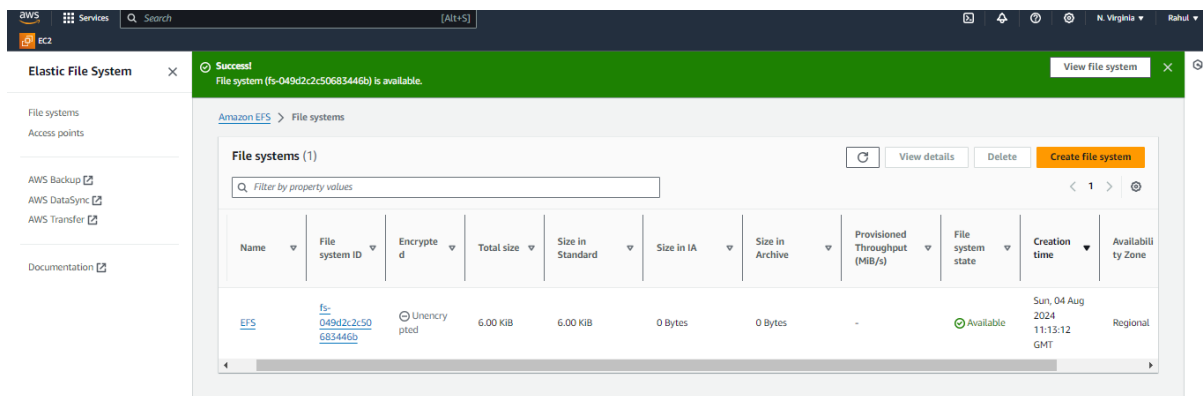
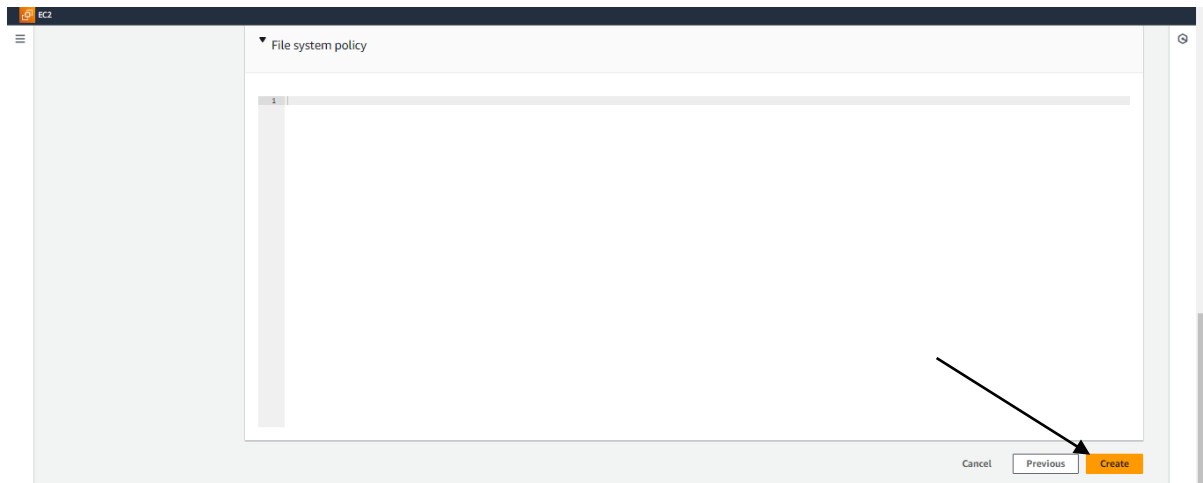
Manual changes will prevent the use of the policy options on the left until the editor is cleared.

Cancel

Previous

Next

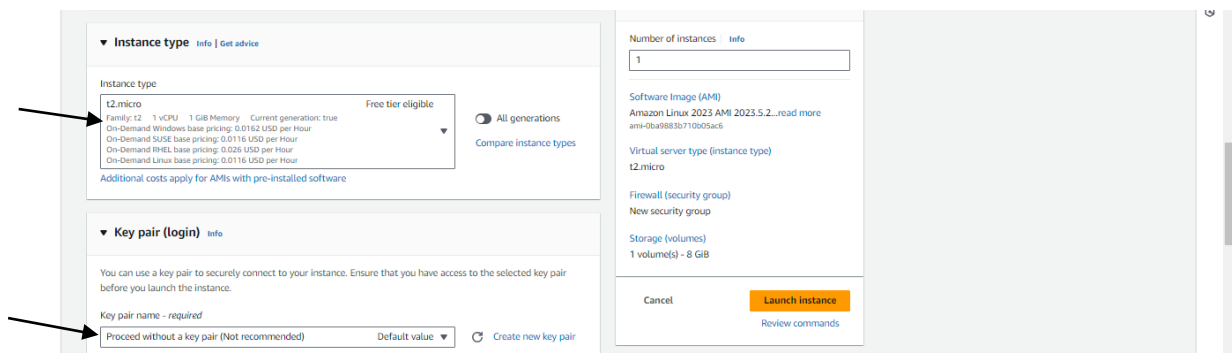
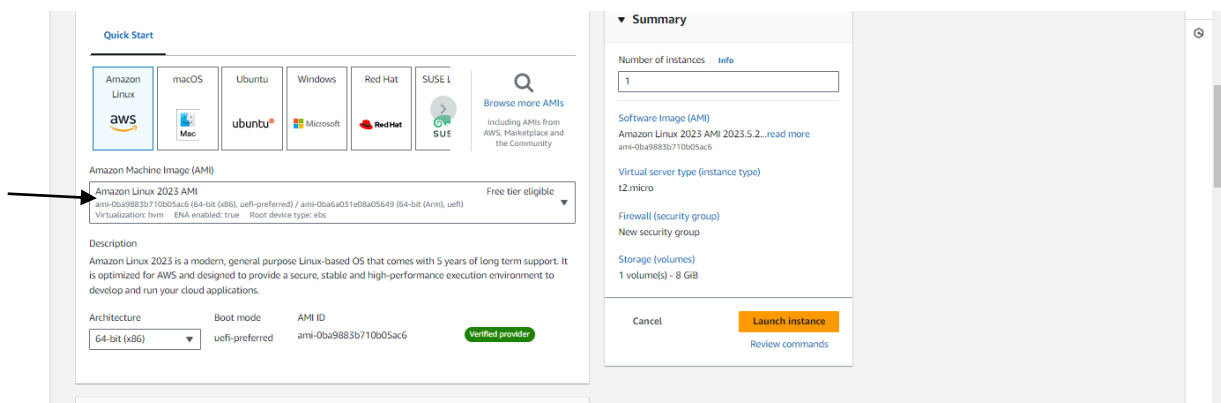
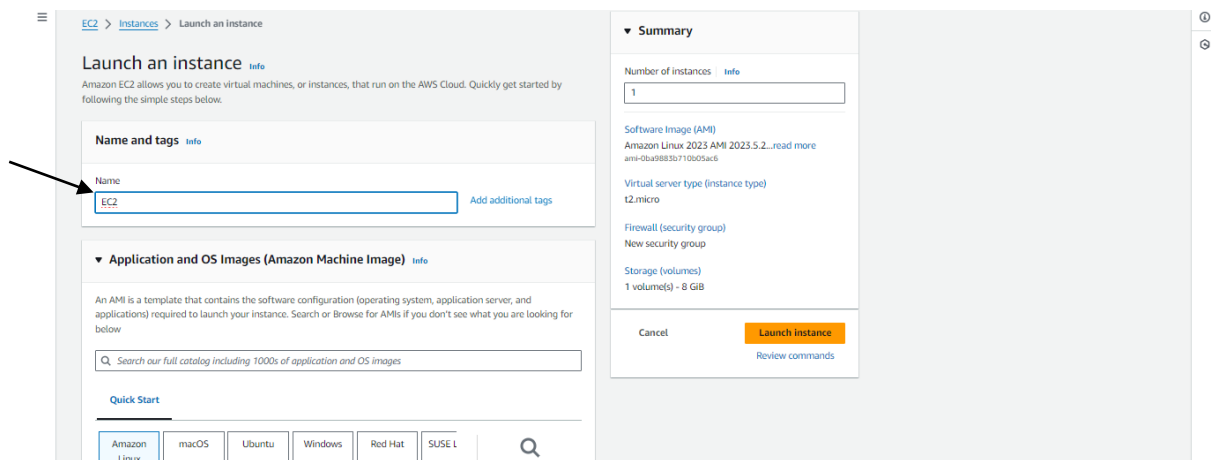
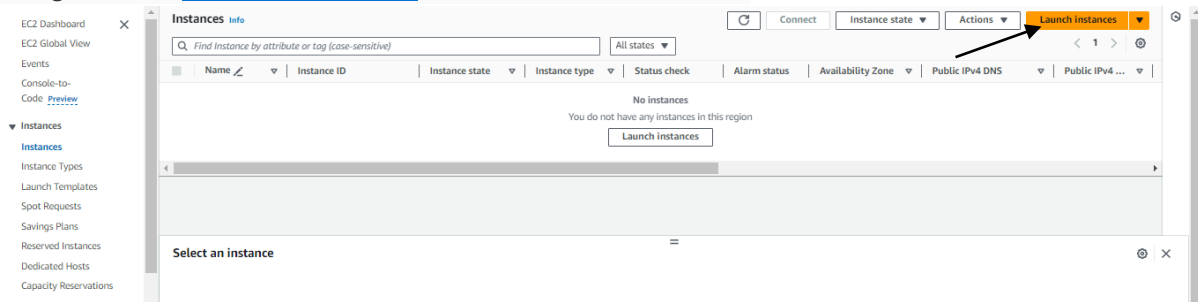
Under Review and create



Successfully Created!!

Step 3: Launch the EC2 instance and mount the file system

Navigate to the [Amazon EC2 console](#) and choose Launch Instance.



▼ Network settings Info

Network Info
vpc-0e9096f0008e45be5

Subnet Info
No preference (Default subnet in any availability zone)

Auto-assign public IP Info
Enable

Additional charges apply when outside of free tier allowance

Edit

Number of instances Info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.5.2...read more
ami-0ba9883b710b05ac6

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Select us-east-1a

▼ Network settings Info

VPC - required Info
vpc-0e9096f0008e45be5 (default) 172.31.0.0/16

Subnet Info
subnet-048ba871409c16025 VPC: vpc-0e9096f0008e45be5 Owner: 014438659001 Availability Zone: us-east-1a IP addresses available: 4090 CIDR: 172.31.16.0/20 Create new subnet

Auto-assign public IP Info
Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☐ Create security group ☒ Select existing security group

Common security groups Info
Select security groups
EC2-sg sg-0f0a1f42310af0e4c VPC: vpc-0e9096f0008e45be5 Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

▼ Summary

Number of instances Info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.5.2...read more
ami-0ba9883b710b05ac6

Virtual server type (instance type)
t2.micro

Firewall (security group)
EC2-sg

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Review commands

Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☐ Create security group ☒ Select existing security group

Common security groups Info
Select security groups
EC2-sg sg-0f0a1f42310af0e4c VPC: vpc-0e9096f0008e45be5 Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Virtual server type (instance type)
t2.micro

Firewall (security group)
EC2-sg

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Review commands

▼ Configure storage Info Advanced

1x 8 GiB gp3 Root volume (Not encrypted)
Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage
Add new volume

Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems Edit

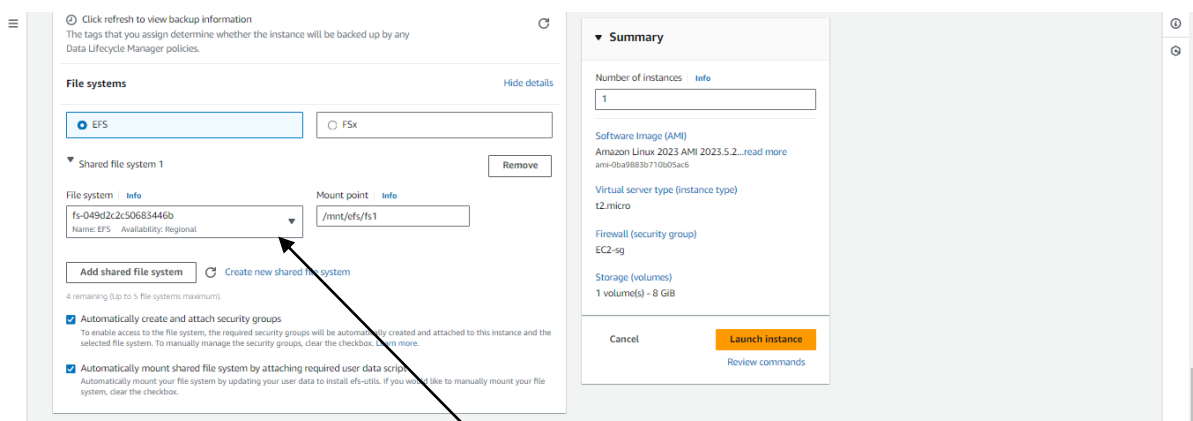
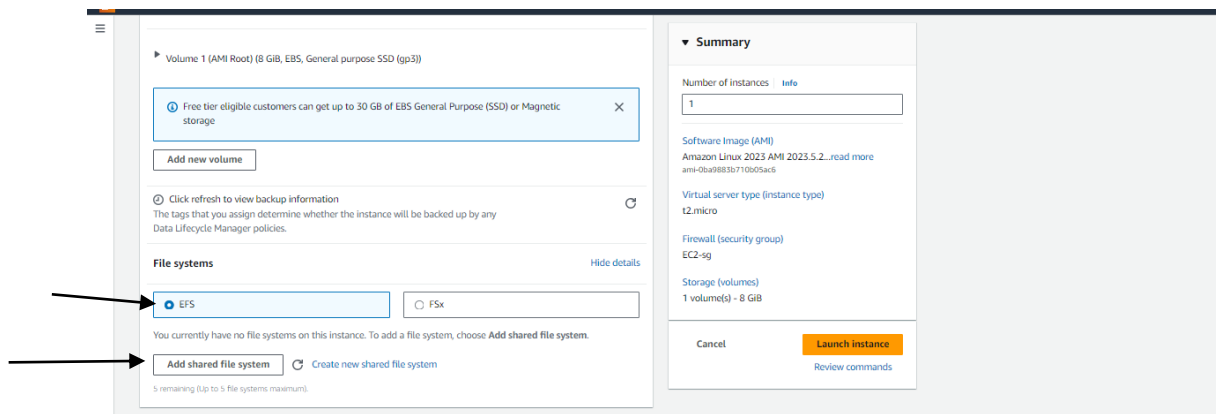
Software Image (AMI)
Amazon Linux 2023 AMI 2023.5.2...read more
ami-0ba9883b710b05ac6

Virtual server type (instance type)
t2.micro

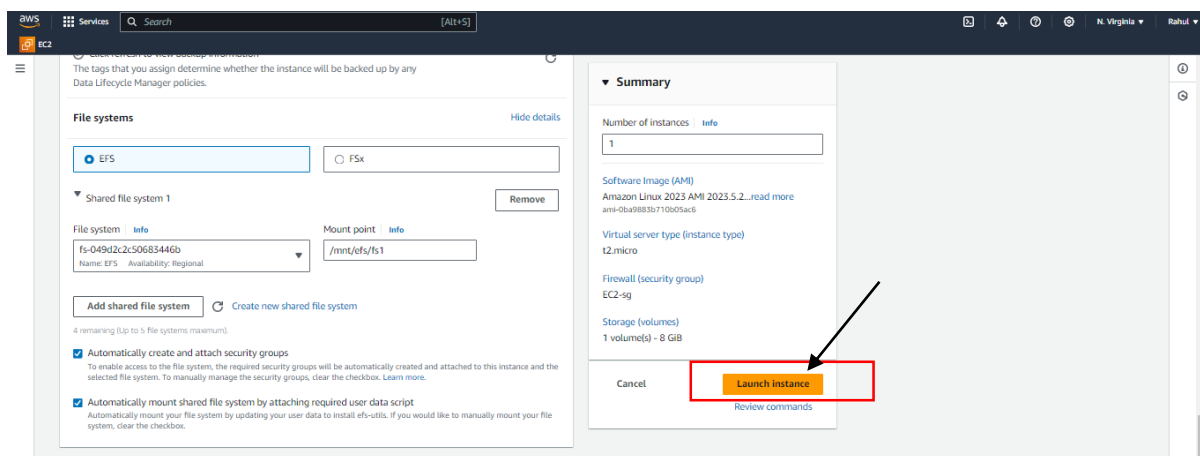
Firewall (security group)
EC2-sg

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Review commands



Select created EFS



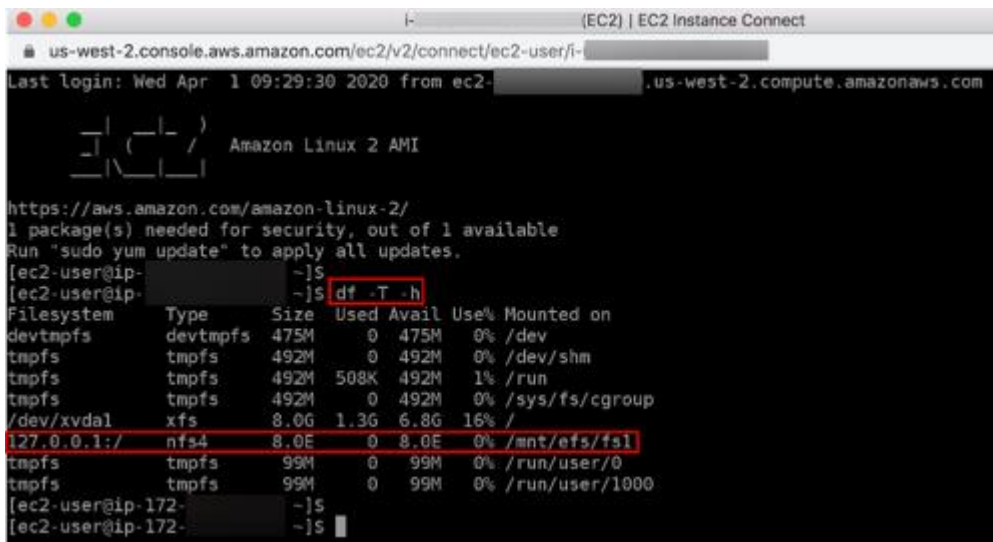
Verify EFS file system is mounted

Navigate to the EC2 console. In the EC2 navigation pane, choose **Instances**, then select the EC2 instance you created in Step 3. Choose **Connect**.

In the Connect to your instance dialog box, choose **EC2 Instance Connect** (browser-based SSH connection) and then choose **Connect**.

type the following command:

`df -T -h`



```
us-west-2.console.aws.amazon.com/ec2/v2/connect/ec2-user/i-  
Last login: Wed Apr 1 09:29:30 2020 from ec2-  
Amazon Linux 2 AMI  
https://aws.amazon.com/amazon-linux-2/  
1 package(s) needed for security, out of 1 available  
Run "sudo yum update" to apply all updates.  
[ec2-user@ip-  
[ec2-user@ip-  
[ec2-user@ip-  
Filesystem      Type      Size      Used Avail Use% Mounted on  
devtmpfs        devtmpfs  475M      0      475M  0% /dev  
tmpfs           tmpfs     492M      0      492M  0% /dev/shm  
tmpfs           tmpfs     492M     508K    492M  1% /run  
tmpfs           tmpfs     492M      0      492M  0% /sys/fs/cgroup  
/dev/xvda1      xfs       8.0G     1.3G     6.8G 16% /  
127.0.0.1:/     nfs4      8.0E      0      8.0E  0% /mnt/efs/fs1  
tmpfs           tmpfs     99M      0      99M  0% /run/user/0  
tmpfs           tmpfs     99M      0      99M  0% /run/user/1000  
[ec2-user@ip-172-  
[ec2-user@ip-172-  
[ec2-user@ip-172-
```

You can see that the EFS File System is mounted at `mnt/efs/fs1` which is the same default path as mentioned in Step 3.4. This verifies that your file system is successfully mounted on the EC2 instance.

Step 5: Clean up

Terminate EC2 instance

5.1 — Open the [Amazon EC2 console](#).

5.2 — In the navigation pane, choose Instances.

5.3 — Select the instance you created for this tutorial, and choose Actions, Instance State, Terminate.

5.4 — Choose Yes, Terminate when prompted for confirmation.

Note: This process can take several seconds to complete. Once your instance has been terminated, the Instance State will change to terminated on your EC2 Console.

Delete EFS file system

5.5 — Open the [Amazon EFS console](#).

5.6 — On the File systems page, select the file system you created for this tutorial.

5.7 — Choose Delete.

5.8 — In the Delete file system dialog box, enter the file system ID shown, and choose Confirm to confirm the delete.

Also Delete Security groups.